INCH-POUND

MIL-PRF-6106/41C
22 March 2012
SUPERSEDING
MIL-PRF-6106/41B
10 November 2000

PERFORMANCE SPECIFICATION SHEET

# RELAY, ELECTROMAGNETIC, 25 AMPERES, 3PNO, WITH 2 AMPERE 1PDT AUXILIARY CONTACTS, HERMETICALLY SEALED, TYPE I

Inactive for new design after 15 November 2002. No superseding

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

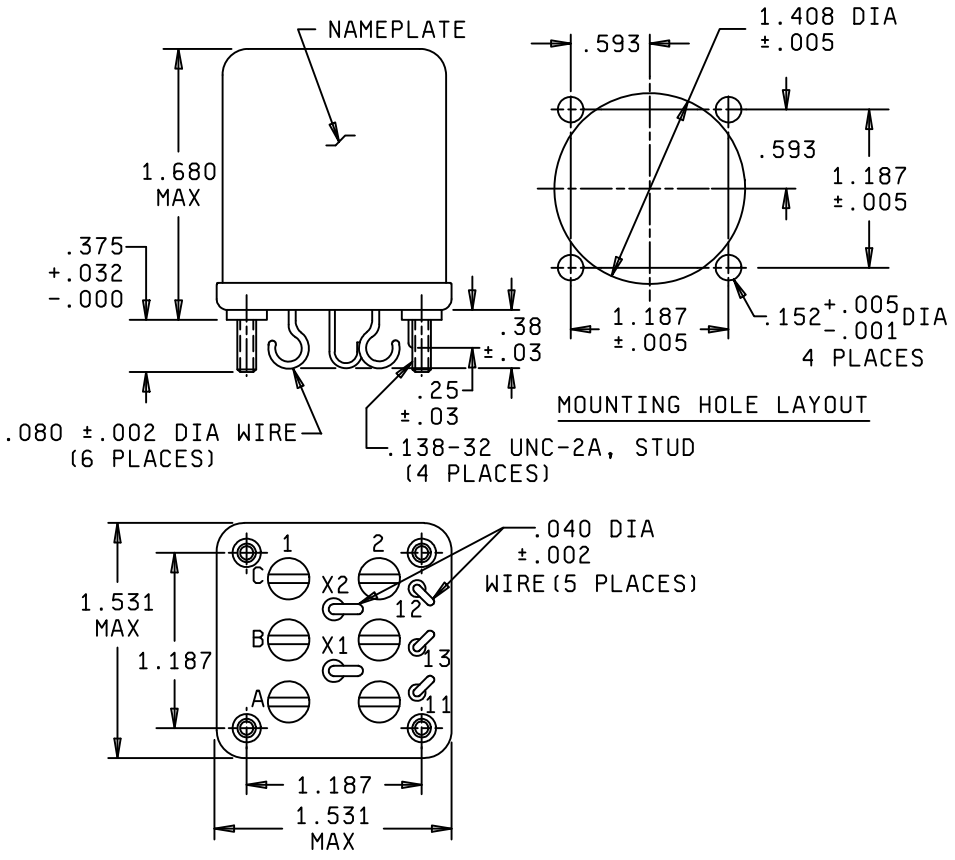
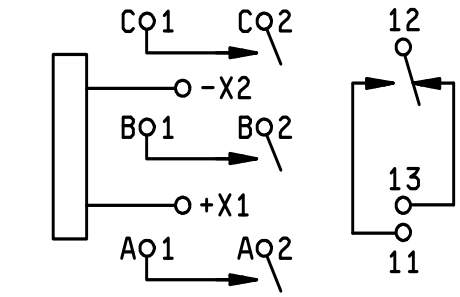


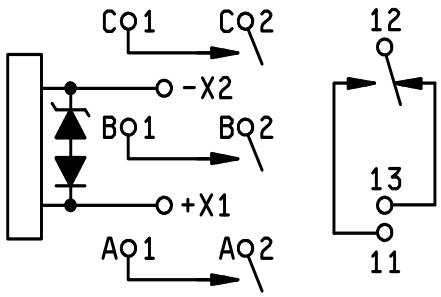

FIGURE 1. Dimensions and configurations.



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CIRCUIT DIAGRAM
DASH NUMBER -001,-002,-003
SEE NOTE 5




CIRCUIT DIAGRAM
DASH NUMBER -004

| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .25 | 6.4 |
| .002 | 0.05 | .375 | 9.53 |
| .005 | 0.13 | .38 | 9.7 |
| .03 | 0.8 | .593 | 15.06 |
| .032 | 0.81 | 1.187 | 30.15 |
| .040 | 1.02 | 1.406 | 35.71 |
| .080 | 2.03 | 1.531 | 38.89 |
| .152 | 3.86 | 1.680 | 42.67 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is .010 (0.25 mm).
4. Terminal numbers need not appear on the relay header provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.

FIGURE 1. Relay, outline drawing - Continued.



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REQUIREMENTS:

Coil data:

Coil data: See table I.

Duty rating: Continuous.

Operational data:

Rated contact load: See table II.

Operate time: See table I.

Release time: See table I.

Maximum induced transient voltage (when specified): 42 V.

Physical:

Dimensions and configurations: See figure 1.

Weight (maximum): 0.44 pound (200 grams).

Terminal type: See figure 1 and table I.

Strength of terminals and mounting studs: Applicable.

Terminal solderability:

Finish: Flat black.

ENVIRONMENTAL CHARACTERISTICS:

Temperature rating: -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

Shock, g-level: 50 g.

Duration: 11 ms.

Maximum duration contact opening: 2 ms.

Vibration - sinusoidal:

Operating:

G-level: 10 g.

Frequency range: 5 to 1,500 Hz.

Nonoperating:

G-level: 15 g.

Frequency range: 20 to 2,000 Hz.

Acceleration: 10 g.



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TABLE I. Mounting, termination, and operating characteristics.

| Dash no. | Coil terminal X1, X2 | Coil data | | | | | | | | | | Time (milliseconds-maximum) | | | | | | Terminals | Mounting |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Nominal | | | Maximum | | Max pickup voltage 1/ | | | Hold voltage 2/ | Drop out voltage 2/ | Operate 3/ | Release 4/ | Contact bounce | | | | | |
| | | | | | | | | | | | | | | Main | | Aux | | | |
| | | Volts 1/ | Freq | Res ±10% | Volts | Amp | Nominal 2/ | High temp test | Cont current test | | | | | NO | NC | NO | NC | | |
| -001 | DC | 28 | DC | --- | 29 | 0.25 | 18 | 19.5 | 22.5 | 7.0 | 1.5 | 20 | 10 | 2.0 | --- | 4.0 | 4.0 | Solder hook | 4 Studs |
| -002 | AC | 115 | 400 | --- | 122 | .055 | 96 | 100 | 108 | 40 | 5.0 | 25 | 50 | 2.0 | --- | 4.0 | 4.0 | Solder hook | 4 Studs |
| -003 | AC | 115 | 50/60 | --- | 122 | .06 | 96 | 100 | 108 | 40 | 5.0 | 25 | 50 | 2.0 | --- | 4.0 | 4.0 | Solder hook | 4 Studs |
| -004 | DC 5/ | 28 | DC | 160 | 29 | 0.25 | 18 | 19.5 | 22.5 | 7.0 | 1.5 | 20 | 10 | 2.0 | --- | 4.0 | 4.0 | Solder hook | 4 Studs |

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ Suppressed dc coil: Back EMF 42 volts maximum.

TABLE II. Rated contact load (amperes per pole) case grounded.

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | | See appropriate footnote |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 50/60 Hz | 400 Hz | 60 Hz | 400 Hz | 50/60 Hz | 400 Hz | 60 HZ | |
| Resistive | 50 | 25 | --- | 2 | 2 | 25 | 25 | 2 | --- | 25 | 25 | --- | --- | --- |
| Inductive | 10 | 15 | --- | 1 | 1 | 25 | 25 | 1 | 1 | 25 | 25 | --- | --- | 2/ |
| Motor | 50 | 20 | --- | --- | --- | 20 | 12 | --- | --- | 20 | 12 | --- | --- | --- |
| Lamp | 50 | 10 | --- | .5 | .5 | 10 | 10 | .5 | --- | 10 | 10 | --- | --- | --- |
| Transfer load | | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | 3/ |
| Mechanical life reduced current | 100 | 6.3 | --- | .5 | .5 | 6.3 | 6.3 | .5 | --- | 6.3 | 6.3 | --- | --- | --- |
| Mixed loads | Applicable per specification | | | | | | | | | | | | | |

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ .7 pF, inductive.
3/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.



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ELECTRICAL CHARACTERISTICS:

Insulation resistance:

Initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric withstanding voltage (sea level):

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,250 V rms | 1,000 V rms |
| Auxiliary contacts | 1,000 V rms | 750 V rms |
| All other points | 1,500 V rms | 1,125 V rms |

Dielectric withstanding voltage (altitude):

| | 80,000 feet |
|---|---|
| Coil to case | 500 V rms |
| Auxiliary contacts | 250 V rms |
| All other points | 500 V rms |

Maximum contact drop:

Initial: 0.150 volt.

After life test: 0.175 volt.

Overload current: 80 amperes dc, 120 amperes ac.

Rupture current: 100 amperes dc, 150 amperes ac.

RFI specification (applicable to coil currents of ac operated relays):

MIL-STD-461.

PART NUMBER: M6106/41 (plus dash number from table I).

Qualification by similarity: See MIL-PRF-6106.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-STD-461

Custodian:
Air Force - 85
DLA-CC

Preparing activity:
DLA-CC

Review activities:
Air Force – 99

(Project 5945-2012-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil/.



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